Introduction to Operations Research, Determinisitic Models

Lecture 5: Sensitivity Analysis and Duality Theory

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Recap: Simplex in matrix form

$$\begin{pmatrix} 1 & \mathbf{c}_B^\top \mathbf{B}^{-1} \\ 0 & \mathbf{B}^{-1} \end{pmatrix} \begin{pmatrix} 1 & -\mathbf{c}^\top & 0 \\ 0 & \mathbf{A} & \mathbf{I} \end{pmatrix} \begin{pmatrix} Z \\ \mathbf{x} \\ \mathbf{x}_S \end{pmatrix} = \begin{pmatrix} \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{b} \\ \mathbf{B}^{-1} \mathbf{b} \end{pmatrix}$$

	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
0	Z	(0)	1	- $\mathbf{c}^\top$	$\mathbf{0}$	0
	$\mathbf{x}_S$		$\mathbf{0}$	$\mathbf{A}$	$\mathbf{I}$	$\mathbf{b}$
$\dots$	$\dots$	$\dots$	$\dots$			
	Z	(0)	1	$\mathbf{y}^* \mathbf{A} - \mathbf{c}^\top$	$\mathbf{y}^*$	$\mathbf{y}^* \mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1} \mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1} \mathbf{b}$

$\mathbf{y}^* \triangleq \mathbf{c}_B^\top \mathbf{B}^{-1}$ denote the coefficients of the slack variable in row (0).

In this lecture, you'll learn

- Sensitivity analysis
- Duality theory

Sensitivity analysis

Sensitivity analysis

After we solve the LP by the simplex method given the constant values of parameters $\mathbf{c}, \mathbf{A}, \mathbf{b}$, what would happen if the parameters take on other possible values?

- does the original optimal solution remain feasible/optimal?
- determine the range of values of the coefficient over which the optimality remains.
- if not optimal, how to find the optimal solution of the new problem utilizing the knowledge of the original LP?

Sensitivity analysis: revised Wyndor example

Original problem

maximize $Z = 3x_1 + 5x_2$

subject to $\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 4 \\ 12 \\ 18 \end{pmatrix}$

$\mathbf{x} \geq \mathbf{0}$

Revised problem

maximize $Z = 4x_1 + 5x_2$

subject to $\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 4 \\ 24 \\ 18 \end{pmatrix}$

$\mathbf{x} \geq \mathbf{0}$

In the original problem, let $\mathbf{c} \triangleq \begin{pmatrix} 3 \\ 5 \end{pmatrix}$, $\mathbf{A} \triangleq \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{pmatrix}$, $\mathbf{b} \triangleq \begin{pmatrix} 4 \\ 12 \\ 18 \end{pmatrix}$.

In the revised problem, let $\bar{\mathbf{c}} \triangleq \begin{pmatrix} 4 \\ 5 \end{pmatrix}$, $\bar{\mathbf{A}} \triangleq \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 2 & 2 \end{pmatrix}$, $\bar{\mathbf{b}} \triangleq \begin{pmatrix} 4 \\ 24 \\ 18 \end{pmatrix}$.

Recall: original Wyndor example

Iteration	Basic variable	row	Coefficient						RHS
			Z	x_1	x_2	x_3	x_4	x_5	
0	Z	(0)	1	-3	-5	0	0	0	0
	x_3	(1)	0	1	0	1	0	0	4
	x_4	(2)	0	0	2	0	1	0	12
	x_5	(3)	0	3	2	0	0	1	18
...	...	...	...						
2	Z	(0)	1	0	0	0	3/2	1	36
	x_3	(1)	0	0	0	1	1/3	-1/3	2
	x_2	(2)	0	0	1	0	1/2	0	6
	x_1	(3)	0	1	0	0	-1/3	1/3	2

In the final tableau, $\mathbf{B} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 2 & 2 \end{pmatrix}$, $\mathbf{B}^{-1} = \begin{pmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & -1/3 & 1/3 \end{pmatrix}$, $\mathbf{y}^* = \begin{pmatrix} 0 \\ 3/2 \\ 1 \end{pmatrix}$.

Sensitivity analysis: revised Wyndor example

Iteration	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
	Z	(0)	1	$\mathbf{y}^*\mathbf{A} - \mathbf{c}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\mathbf{b}$

With the same row operations, we obtain the final tableau for the revised problem.

Iteration	Basic variable	row	Coefficient						RHS
			Z	x_1	x_2	x_3	x_4	x_5	
0	Z	(0)	1	-4	-5	0	0	0	0
	x_3	(1)	0	1	0	1	0	0	4
	x_4	(2)	0	0	2	0	1	0	24
	x_5	(3)	0	2	2	0	0	1	18
...	.....	...							
2	Z	(0)	1	-2	0	0	3/2	1	36
	x_3	(1)	0	1/3	0	1	1/3	-1/3	2
	x_2	(2)	0	0	1	0	1/2	0	6
	x_1	(3)	0	2/3	0	0	-1/3	1/3	2

Sensitivity analysis: general procedure

After we solve the LP by the simplex method given the constant values of parameters $(\mathbf{c}, \mathbf{A}, \mathbf{b})$, what would be the optimal solutions if some of the parameters take on other possible values?

- **Revision of model:** Make the desired change or changes in the model to be investigated.
- **Revision of final tableau:** Use the simplex in matrix form to determine the resulting changes in the final simplex tableau after the same row operations.
- **Conversion to proper form** of simplex tableau from Gaussian elimination

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- **Conversion to proper form** of simplex tableau from Gaussian elimination

- **Feasibility test:** checking whether all values in the RHS column (basic variable values) of the tableau are nonnegative.

- **Optimality test:** checking whether all its nonbasic variable coefficients in row 0 of the tableau still are nonnegative.

- **Reoptimization:** If this solution fails either test, the new optimal solution can be obtained by using the current tableau from step 3 as the initial simplex tableau with any necessary conversions.

Sensitivity analysis

After we solve the LP by the simplex method given the constant values of parameters $(\mathbf{c}, \mathbf{A}, \mathbf{b})$, what would happen if some of the parameters take on other possible values?

- change in RHS
- change in coefficient of NBV
- introduce of a new variable
- changes in coefficient of BV
- add a new constraint

Sensitivity analysis: revised Wyndor example 1

Original problem

$$\begin{aligned} \text{maximize} \quad & Z = 3x_1 + 5x_2 \\ \text{subject to} \quad & \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 4 \\ 12 \\ 18 \end{pmatrix} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

Revised problem

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Suppose we have obtained the final tableau for the original Wyndor example.

	Basic variable	row	Coefficient						RHS
			Z	x_1	x_2	x_3	x_4	x_5	
Final tableau	Z	(0)	1	0	0	0	3/2	1	36
	x_3	(1)	0	0	0	1	1/3	-1/3	2
	x_2	(2)	0	0	1	0	1/2	0	6
	x_1	(3)	0	1	0	0	-1/3	1/3	2

Case 1: changes in RHS

- Case 1: only $\mathbf{b}$ changes to $\bar{\mathbf{b}}$, while $\mathbf{A}$ and $\mathbf{c}$ remain the same.

Given the final simplex tableau for the original problem, we can obtain the following tableau with the same row operations (matrix multiplication).

	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
	Z	(0)	1	$\mathbf{y}^*\mathbf{A} - \mathbf{c}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\bar{\mathbf{b}}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\bar{\mathbf{b}}$

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Observe: in the **final revised simplex tableau**, the basic variables remain unchanged (no further Gaussian elimination is needed), the coefficients of all non-basic variables in row (0) are nonnegative. So <u>if the current basic solution is feasible, then it is optimal.</u>

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if the new RHS $\mathbf{B}^{-1}\bar{\mathbf{b}} \geq \mathbf{0}$, then the current basic solution is feasible and thus optimal.

if only b_j changes by Δb_j, then to remain feasible, we must have
$\mathbf{B}^{-1}(\mathbf{b} + \Delta\mathbf{b}) = \mathbf{B}^{-1}\mathbf{b} + (\mathbf{B}^{-1})_j \Delta b_j \geq \mathbf{0} \implies (\mathbf{B}^{-1})_j \Delta b_j \geq -\mathbf{B}^{-1}\mathbf{b}$

Sensitivity analysis: revised Wyndor example 2

revised Wyndor example 1

maximize $Z = 3x_1 + 5x_2$

subject to $\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 4 \\ 24 \\ 18 \end{pmatrix}$

$\mathbf{x} \geq \mathbf{0}$

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subject to $\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 4 \\ 24 \\ 18 \end{pmatrix}$

$\mathbf{x} \geq \mathbf{0}$

Suppose we have obtained the final tableau for the revised Wyndor example 1.

	Basic variable	row	Coefficient						RHS
			Z	x_1	x_2	x_3	x_4	x_5	
Final	Z	(0)	1	9/2	0	0	0	5/2	45
	x_3	(1)	0	1	0	1	0	0	4
	x_2	(2)	0	3/2	1	0	0	1/2	9
	x_4	(3)	0	-3	0	0	1	-1	6

Case 2a: changes in coefficients of NBV

Given the final simplex tableau for the original problem, we can obtain the following tableau with the same row operations (matrix multiplication)

	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
	Z	(0)	1	$\mathbf{y}^*\bar{\mathbf{A}} - \bar{\mathbf{c}}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\bar{\mathbf{A}}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\mathbf{b}$

- Case 2a: only $\mathbf{A}_j$ and $\mathbf{c}_j$ corresponding to a nonbasic variable x_j for some $j \in [n]$ change to $\bar{\mathbf{A}}_j$ and $\bar{\mathbf{c}}_j$ respectively, while $\mathbf{b}$ remains the same.

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Given the final simplex tableau for the original problem, we can obtain the following tableau with the same row operations (matrix multiplication)

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	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\bar{\mathbf{A}}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\mathbf{b}$

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Observe: in the **final revised simplex tableau**, the coefficients of basic variables remain the same (no further Gaussian elimination is needed), the RHS remains the same (the original optimal solution is feasible).

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	Z	(0)	1	$\mathbf{y}^*\bar{\mathbf{A}} - \bar{\mathbf{c}}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
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Observe: in the **final revised simplex tableau**, the coefficients of basic variables remain the same (no further Gaussian elimination is needed), the RHS remains the same (the original optimal solution is feasible).

if $\mathbf{y}^*\bar{\mathbf{A}}_j - \bar{c}_j \geq \mathbf{0}$, and $\mathbf{y}^* \geq \mathbf{0}$, then the the current BF solution is optimal to the revised problem.

if only c_j changes by Δc_j ($\bar{c}_j = c_j + \Delta c_j$), then to remain optimal, we must have $\Delta c_j \leq \mathbf{y}^*\mathbf{A}_j - c_j$, which is the maximum increase in c_j to keep the current BF solution

Case 2b: introduction of a new variable

Suppose that we obtained the final simplex tableau for the original problem.

Iteration	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
	Z	(0)	1	$\mathbf{y}^*\mathbf{A} - \mathbf{c}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\mathbf{b}$

• Case 2b: a new variable x_j is introduced with the appropriate coefficients into the objective function and constraints of the current model.

Case 2b: introduction of a new variable

Suppose that we obtained the final simplex tableau for the original problem.

Iteration	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
	Z	(0)	1	$\mathbf{y}^*\mathbf{A} - \mathbf{c}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\mathbf{b}$

• Case 2b: a new variable x_j is introduced with the appropriate coefficients into the objective function and constraints of the current model.

Observe: We pretend that the new variable x_j actually was in the original model with all its coefficients equal to zero (so that they still are zero in the final simplex tableau) and that x_j is a nonbasic variable in the current BF solution.

So we could repeat the optimality check in Case 2a.

Sensitivity analysis: revised Wyndor example 3

revised Wyndor example 1

maximize $Z = 3x_1 + 5x_2$

subject to $\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 4 \\ 24 \\ 18 \end{pmatrix}$

$\mathbf{x} \geq \mathbf{0}$

revised Wyndor example 3

maximize $Z = 3x_1 + 3x_2$

subject to $\begin{pmatrix} 1 & 0 \\ 0 & 3 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 4 \\ 24 \\ 18 \end{pmatrix}$

$\mathbf{x} \geq \mathbf{0}$

Suppose we have obtained the final tableau for the revised Wyndor example 1.

	Basic variable	row	Coefficient						RHS
			Z	x_1	x_2	x_3	x_4	x_5	
Final	Z	(0)	1	9/2	0	0	0	5/2	45
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	x_2	(2)	0	3/2	1	0	0	1/2	9
	x_4	(3)	0	-3	0	0	1	-1	6

Sensitivity analysis: revised Wyndor example 3

	Basic variable	row	Coefficient						RHS
			Z	x_1	x_2	x_3	x_4	x_5	
Revised final tableau	Z	(0)	1	9/2	7	0	0	5/2	45
	x_3	(1)	0	1	0	1	0	0	4
	x_2	(2)	0	3/2	2	0	0	1/2	9
	x_4	(3)	0	-3	-1	0	1	-1	6
Convert into the proper form									
New final tableau									

Convert the revised final tableau to the proper form, and then proceed with the simplex method to obtain the new optimal solution.

Case 3: changes in coefficients of BV

Given the final simplex tableau for the original problem, we can obtain the following tableau with the same row operations (matrix multiplication)

	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
	Z	(0)	1	$\mathbf{y}^*\bar{\mathbf{A}} - \bar{\mathbf{c}}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
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- Case 3: only $\mathbf{A}_j$ and $\mathbf{c}_j$ corresponding to a basic variable x_j for some $j \in [n]$ change to $\bar{\mathbf{A}}_j$ and $\bar{\mathbf{c}}_j$ respectively, while $\mathbf{b}$ remains the same.

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Observe: in the **final revised simplex tableau**, the coefficients of the basic variable x_j changes (further Gaussian elimination is needed).

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Observe: in the **final revised simplex tableau**, the coefficients of the basic variable x_j changes (further Gaussian elimination is needed).

For the new basic solution after Gaussian elimination, check the feasibility and optimality.

if only c_j changes by Δc_j, we can compute the range of Δc_j such that the coefficients in row (0) remains nonnegative and thus the current BF solution remains optimal.

Case 4: add a new constraint

First, check whether the current BF satisfies the new constraint.

If the current BF satisfies the new constraint, then the current BF solution is feasible and thus optimal.

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If the current BF satisfies the new constraint, then the current BF solution is feasible and thus optimal.

If the current BF violates the new constraint,

- introduce this constraint into the final simplex tableau (as an additional row) just as if this were the initial tableau, where the usual additional variable (slack variable or artificial variable) is designated to be the basic variable for this new row.
- apply the conversion to proper form from Gaussian elimination, and then proceed with Simplex procedure.

Duality theory

Dual problem

Primal problem (general form)

$$\begin{array}{ll} \textbf{maximize} & Z = \mathbf{c}^\top \mathbf{x} \\ \text{subject to} & \mathbf{A}\mathbf{x} \le \mathbf{b} \\ & \mathbf{x}(\in \mathbb{R}^n) \ge 0 \end{array}$$

Dual problem

$$\begin{array}{ll} \textbf{minimize} & W = \mathbf{b}^\top \mathbf{y} \\ \text{subject to} & \mathbf{A}^\top \mathbf{y} \ge \mathbf{c} \\ & \mathbf{y}(\in \mathbb{R}^m) \ge 0 \end{array}$$

Let $\mathbf{A} \triangleq (\mathbf{A}_1, \ldots, \mathbf{A}_n) = \begin{pmatrix} \mathbf{a}_1^\top \\ \vdots \\ \mathbf{a}_m^\top \end{pmatrix}$. The correspondence of primal and dual problem is:

primal problem	**dual problem**
Constraint $\mathbf{a}_i^\top \mathbf{x} \le b_i$	Variable $\mathbf{y}_i \ge 0$
Variable $\mathbf{x}_j \ge 0$	Constraint $\mathbf{A}_j^\top \mathbf{y} \ge c_j$
coefficient in objective	RHS
RHS	coefficient in objective

Dual problem

Primal problem

maximize $Z = 3x_1 + 5x_2$

subject to $\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 4 \\ 12 \\ 18 \end{pmatrix}$

$\mathbf{x} \geq \mathbf{0}$

Dual problem

minimize $W = 4y_1 + 12y_2 + 18y_3$

subject to $\begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \geq \begin{pmatrix} 3 \\ 5 \end{pmatrix}$

$\mathbf{y} \geq \mathbf{0}$

Weak duality property

Weak duality

If $\mathbf{x}$ is a feasible solution for the primal problem and $\mathbf{y}$ is a feasible solution for the dual problem, then $\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}$

Proof. Since $\mathbf{A}\mathbf{x} \leq \mathbf{b}$, we have $\mathbf{y}^\top \mathbf{A}\mathbf{x} \leq \mathbf{y}^\top \mathbf{b}$. On the other hand, since $\mathbf{A}^\top \mathbf{y} \geq \mathbf{c}$, we have $\mathbf{x}^\top \mathbf{A}^\top \mathbf{y} \geq \mathbf{x}^\top \mathbf{c}$. Combining the two inequality, we thus obtain that $\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}$.

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Corollary 1

Let $\bar{\mathbf{x}}$ be a feasible solution for the primal problem and $\bar{\mathbf{y}}$ be a feasible solution for the dual problem. If $\mathbf{c}^\top \bar{\mathbf{x}} = \mathbf{b}^\top \bar{\mathbf{y}}$, then $\bar{\mathbf{x}}$ is optimal for the primal and $\bar{\mathbf{y}}$ is optimal for the dual.

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Proof. Since $\mathbf{A}\mathbf{x} \leq \mathbf{b}$, we have $\mathbf{y}^\top \mathbf{A}\mathbf{x} \leq \mathbf{y}^\top \mathbf{b}$. On the other hand, since $\mathbf{A}^\top \mathbf{y} \geq \mathbf{c}$, we have $\mathbf{x}^\top \mathbf{A}^\top \mathbf{y} \geq \mathbf{x}^\top \mathbf{c}$. Combining the two inequality, we thus obtain that $\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}$.

Corollary 1

Let $\bar{\mathbf{x}}$ be a feasible solution for the primal problem and $\bar{\mathbf{y}}$ be a feasible solution for the dual problem. If $\mathbf{c}^\top \bar{\mathbf{x}} = \mathbf{b}^\top \bar{\mathbf{y}}$, then $\bar{\mathbf{x}}$ is optimal for the primal and $\bar{\mathbf{y}}$ is optimal for the dual.

A maximization (minimization) problem is unbounded if its optimal value is $+\infty$ $(-\infty)$.

Corollary 2

If one problem has feasible solutions and an unbounded objective function (and so no optimal solution), then the other problem has no feasible solutions.

Strong duality property

Strong duality

If $\mathbf{x}^*$ is an optimal solution for the primal problem of LP and $\mathbf{y}^*$ is an optimal solution for its dual problem, then $\mathbf{c}^\top \mathbf{x}^* = \mathbf{b}^\top \mathbf{y}^*$.

Primal problem

$$\begin{array}{ll} \textbf{maximize} & Z = \mathbf{c}^\top \mathbf{x} \\ \text{subject to} & \mathbf{A}\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x}(\in \mathbb{R}^n) \geq 0 \end{array}$$

Dual problem

$$\begin{array}{ll} \textbf{minimize} & Z = \mathbf{b}^\top \mathbf{y} \\ \text{subject to} & \mathbf{A}^\top \mathbf{y} \geq \mathbf{c} \\ & \mathbf{y}(\in \mathbb{R}^m) \geq 0 \end{array}$$

Final tableau of the simplex method:

	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
Final	Z	(0)	1	$\mathbf{y}^*\mathbf{A} - \mathbf{c}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\mathbf{b}$

Adaptation to the non-standard form

For a non-standard LP, we can reformulate it as a standard LP and construct the dual problem.

Primal	**Dual**
Maximize Z	Minimize W
Constraint i: $\leq$ form	Variable $y_i \geq 0$
$=$ form	y_i unconstrained
$\geq$ form	$y_i \leq 0$
Variable $x_j \geq 0$	Constraint j: $\geq$ form
x_j unconstrained	$=$ form
$x_j \leq 0$	$\leq$ form

Adaptation to the non-general form

Dual of the dual problem is the primal problem.

Primal problem (general form)

$$\boxed{\begin{array}{ll} \textbf{maximize} & Z = \mathbf{c}^\top \mathbf{x} \\ \text{subject to} & \mathbf{A}\mathbf{x} \le \mathbf{b} \\ & \mathbf{x} \ge 0 \end{array}}$$

$\Updownarrow$

dual of the dual

$$\boxed{\begin{array}{ll} \textbf{minimize} & (-Z) = -\mathbf{c}^\top \mathbf{x} \\ \text{subject to} & -\mathbf{A}\mathbf{x} \ge -\mathbf{b} \\ & \mathbf{x} \ge 0 \end{array}}$$

$\Rightarrow$

Dual problem

$$\boxed{\begin{array}{ll} \textbf{minimize} & W = \mathbf{b}^\top \mathbf{y} \\ \text{subject to} & \mathbf{A}^\top \mathbf{y} \ge \mathbf{c} \\ & \mathbf{y} \ge 0 \end{array}}$$

$\Updownarrow$

converted to general form

$$\boxed{\begin{array}{ll} \textbf{maximize} & (-W) = -\mathbf{b}^\top \mathbf{y} \\ \text{subject to} & -\mathbf{A}^\top \mathbf{y} \le -\mathbf{c} \\ & \mathbf{y} \ge 0 \end{array}}$$

$\Leftarrow$

The relationships of the dual problem to the primal problem also hold in reverse

Duality theorem

Duality theorem

The following are the only possible relationships between the primal and dual problems.

1. If one problem has feasible solutions and a bounded objective function (and so has an optimal solution), then so does the other problem. Moreover, both the weak and strong duality properties hold.
2. If one problem has feasible solutions and an unbounded objective function (and so no optimal solution), then the other problem has no feasible solutions.
3. If one problem has no feasible solutions, then the other problem has either no feasible solutions or an unbounded objective function.

Complementary solution property

	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
Some	Z	(0)	1	$\mathbf{y}^*\mathbf{A} - \mathbf{c}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\mathbf{b}$

Complementary solutions property: At each iteration, the simplex method simultaneously identifies a CPF solution $\mathbf{x}$ for the primal problem and a complementary solution $\mathbf{y}$ for the dual problem (found in row 0, the coefficients of the slack variables), where $\mathbf{c}^\top\mathbf{x} = \mathbf{b}^\top\mathbf{y}$. If $\mathbf{x}$ is not optimal for the primal problem, then $\mathbf{y}$ is not feasible for the dual problem.

Complementary solution property

	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
Some	Z	(0)	1	$\mathbf{y}^*\mathbf{A} - \mathbf{c}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\mathbf{b}$

Complementary solutions property: At each iteration, the simplex method simultaneously identifies a CPF solution $\mathbf{x}$ for the primal problem and a complementary solution $\mathbf{y}$ for the dual problem (found in row 0, the coefficients of the slack variables), where $\mathbf{c}^\top\mathbf{x} = \mathbf{b}^\top\mathbf{y}$. If $\mathbf{x}$ is not optimal for the primal problem, then $\mathbf{y}$ is not feasible for the dual problem.

Complementary optimal solutions property: The simplex method simultaneously identifies an optimal solution $\mathbf{x}^*$ for the primal problem and a complementary optimal solution $\mathbf{y}^*$ for the dual problem (found in row 0, the coefficients of the slack variables), where $\mathbf{c}^\top\mathbf{x}^* = \mathbf{b}^\top\mathbf{y}^*$.

Complementary solution property

	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
Some	Z	(0)	1	$\mathbf{y}^*\mathbf{A} - \mathbf{c}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\mathbf{b}$

Complementary solutions property: At each iteration, the simplex method simultaneously identifies a CPF solution $\mathbf{x}$ for the primal problem and a complementary solution $\mathbf{y}$ for the dual problem (found in row 0, the coefficients of the slack variables), where $\mathbf{c}^\top\mathbf{x} = \mathbf{b}^\top\mathbf{y}$. If $\mathbf{x}$ is not optimal for the primal problem, then $\mathbf{y}$ is not feasible for the dual problem.

Complementary optimal solutions property: The simplex method simultaneously identifies an optimal solution $\mathbf{x}^*$ for the primal problem and a complementary optimal solution $\mathbf{y}^*$ for the dual problem (found in row 0, the coefficients of the slack variables), where $\mathbf{c}^\top\mathbf{x}^* = \mathbf{b}^\top\mathbf{y}^*$.

Symmetry property: For any primal problem and its dual problem, all relationships between them must be symmetric because the dual of this dual problem is this primal problem.

Complementary basic solution property

	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
Some	Z	(0)	1	$\mathbf{y}^*\mathbf{A} - \mathbf{c}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\mathbf{b}$

Complementary basic solutions property:

• Each basic solution (feasible or not) in the primal problem has a complementary basic solution (feasible or not) in the dual problem, with the same objective function values. Given row 0 of the simplex tableau for the primal basic solution, the complementary dual basic solution is $\boxed{(\mathbf{y}^*, \mathbf{y}^*\mathbf{A} - \mathbf{c}^\top)}$.

Complementary basic solution property

	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
Some	Z	(0)	1	$\mathbf{y}^*\mathbf{A} - \mathbf{c}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\mathbf{b}$

Complementary basic solutions property:

• Each basic solution (feasible or not) in the primal problem has a complementary basic solution (feasible or not) in the dual problem, with the same objective function values. Given row 0 of the simplex tableau for the primal basic solution, the complementary dual basic solution is $\boxed{(\mathbf{y}^*, \mathbf{y}^*\mathbf{A} - \mathbf{c}^\top)}$.

• If the basic solution is not optimal for the primal problem, then the complementary dual basic solution is not feasible for the dual problem.

Complementary basic solution property

	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
Some	Z	(0)	1	$\mathbf{y}^*\mathbf{A} - \mathbf{c}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\mathbf{b}$

Complementary basic solutions property:

• Each basic solution (feasible or not) in the primal problem has a complementary basic solution (feasible or not) in the dual problem, with the same objective function values. Given row 0 of the simplex tableau for the primal basic solution, the complementary dual basic solution is $\boxed{(\mathbf{y}^*, \mathbf{y}^*\mathbf{A} - \mathbf{c}^\top)}$.

• If the basic solution is not optimal for the primal problem, then the complementary dual basic solution is not feasible for the dual problem.

Complementary optimal solutions property: An optimal basic solution in the primal problem has a complementary optimal basic solution in the dual problem, with the same objective function values.

Duality and sensitivity analysis

Duality and sensitivity analysis of NBV

• Case 2a: only $\mathbf{A}_j$ and $\mathbf{c}_j$ corresponding to a nonbasic variable x_j for some $j \in [n]$ change to $\bar{\mathbf{A}}_j$ and $\bar{\mathbf{c}}_j$ respectively, while $\mathbf{b}$ remains the same.

Observe:

1. the original optimal BF solution remains feasible.

2. if $\mathbf{y}^* \bar{\mathbf{A}}_j - \bar{c}_j \geq \mathbf{0}$, and $\mathbf{y}^* \geq \mathbf{0}$, then the the current BF solution is optimal to the revised problem.

Interpretation from the dual: checking whether this complementary basic dual solution still satisfies this revised constraint. If it is satisfied, then the complementary basic dual solution is feasible and thus the original BF optimal solution is optimal.

Duality and shadow price

We define the **shadow price** for the ith constraint of an LP to be the amount by which the optimal objective value is improved if the right hand side of the ith constraint is increased by 1.

- Case 1: only $\mathbf{b}_i$ changes to $\mathbf{b}_i + 1$, while other parameters remain the same.

	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
	Z	(0)	1	$\mathbf{y}^*\mathbf{A} - \mathbf{c}$	$\mathbf{y}^*$	$\mathbf{y}^*(\mathbf{b} + \mathbf{e}_i)$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}(\mathbf{b} + \mathbf{e}_i)$

where $\mathbf{e}_i = (0, \ldots, 0, 1, 0, \ldots, 0)$.

Observe:

1. if $\mathbf{B}^{-1}(\mathbf{b} + \mathbf{e}_i) \geq \mathbf{0}$, then the original optimal solution is feasible and thus optimal.

2 Given the new RHS $\mathbf{B}^{-1}(\mathbf{b} + \mathbf{e}_i) \geq \mathbf{0}$, the shadow price is the optimal dual solution $\mathbf{y}_i^*$.

3. For a maximization problem, the shadow price corresponding to a $\leq$ functional constraint can be interpreted as the maximum amount that the company would pay for extra unit of the resource associated with the ith constraint.

Shadow price for non-general primal problem

For a maximization primal problem, the shadow price is the optimal dual solution $\mathbf{y}_i^*$.

For a $\leq$ functional constraint ($\mathbf{a}_i^\top \mathbf{x} \leq b_i$), the shadow price is nonnegative ($y_i^* \geq 0$)

For a $\geq$ functional constraint ($\mathbf{a}_i^\top \mathbf{x} \geq b_i$), the shadow price is nonpositive ($y_i^* \leq 0$)

For a $=$ functional constraint ($\mathbf{a}_i^\top \mathbf{x} = b_i$), the shadow price is undetermined.

Complementary slackness property

Complementary slackness property

	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
Some	Z	(0)	1	$\mathbf{y}^*\mathbf{A} - \mathbf{c}$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\mathbf{b}$

Observation: every primal decision variable x_j is associated with the dual surplus variable $y_{e,j}$, every primal slack variable $x_{S,i}$ is associated with the dual decision variable y_i.

Complementary slackness property: For the primal basic solution $(\mathbf{x}, \mathbf{x}_S)$ and complementary dual basic solution $(\mathbf{y}, \mathbf{y}_e)$, the basic variables in the primal basic solution (feasible or not) are associated with nonbasic variables in the complementary dual basic solution (feasible or not). The nonbasic variables in the primal basic solution are associated with basic variables in the complementary dual basic solution.

Equivalently, **complementary slackness property** claims that for the primal basic solution $(\mathbf{x}, \mathbf{x}_S)$ and complementary dual basic solution $(\mathbf{y}, \mathbf{y}_e)$, we must have $\mathbf{x}_{S,i}\,\mathbf{y}_i = 0$, and $\mathbf{y}_{e,j}\,\mathbf{x}_j = 0$ for every $i \in [m]$, and $j \in [n]$.

Complementary slackness property

Complementary slackness property can help us to identify the complementary dual basic solution given the primal solution.

Primal problem

maximize $Z = 3x_1 + 5x_2$

subject to $\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 4 \\ 12 \\ 18 \end{pmatrix}$

$\mathbf{x} \geq \mathbf{0}$

Dual problem

minimize $W = 4y_1 + 12y_2 + 18y_3$

subject to $\begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \geq \begin{pmatrix} 3 \\ 5 \end{pmatrix}$

$\mathbf{y} \geq \mathbf{0}$

Compute the complementary dual basic solution:

- (0,0,4,12,18) & (0,0,0,-3,-5)
- (4,0,0,12,6) & (3,0,0,0,-5).
- (6,0,-2,12,0) & (0,0,1,0,-3)

Complementary slackness theorem

Complementary slackness theorem

For a **feasible** primal solution $\mathbf{x}$, let $\mathbf{x}_s$ be the slack variable for the primal problem. For a **feasible** dual solution $\mathbf{y}$, let $\mathbf{y}_e$ be the excess (surplus) variable for the dual problem. Then

$$\mathbf{x} \text{ is primal optimal and } \mathbf{y} \text{ is dual optimal}$$

$$\iff \boxed{\mathbf{y}\,\mathbf{x}_s = 0, \text{ and } \mathbf{y}_e\,\mathbf{x} = 0} \iff \boxed{\mathbf{x}_{S,i}\,\mathbf{y}_i = 0, \text{ and } \mathbf{y}_{e,j}\,\mathbf{x}_j = 0, \forall\, i \in [m], \text{ and } j \in [n]}.$$

Complementary slackness theorem

Complementary slackness theorem

For a **feasible** primal solution $\mathbf{x}$, let $\mathbf{x}_s$ be the slack variable for the primal problem. For a **feasible** dual solution $\mathbf{y}$, let $\mathbf{y}_e$ be the excess (surplus) variable for the dual problem. Then

$\mathbf{x}$ is primal optimal and $\mathbf{y}$ is dual optimal

$$\iff \boxed{\mathbf{y}\,\mathbf{x}_s = 0, \text{ and } \mathbf{y}_e\,\mathbf{x} = 0} \iff \boxed{\mathbf{x}_{S,i}\,\mathbf{y}_i = 0, \text{ and } \mathbf{y}_{e,j}\,\mathbf{x}_j = 0, \forall\, i \in [m], \text{ and } j \in [n]}.$$

Interpretation:

• If a constraint in the primal problem is inactive (non-binding) at a primal feasible solution ($\mathbf{x}_{S,i} > 0$), then the corresponding dual variable y_i must equal 0.

• At a primal optimal solution, if a constraint in the primal problem is inactive ($\mathbf{x}_{S,i} > 0$), meaning that there are extra resources that are not used at optimality, then extra resources should be worthless and the corresponding shadow price $\mathbf{y}_i$ must equal 0.

• At a dual optimal solution, if a dual variable $\mathbf{y}_i > 0$, meaning that the shadow price is positive and extra resources are valuable, then the corresponding slack variable equals 0 and the corresponding primal constraint should be active.

Reading materials

Section 4.6 - 4.8, 5.1 - 5.3, Section 6.1-6.4, 7.1 - 7.2, *Introduction to Operations Research* by Hillier and Lieberman